4 Data integration and management

This chapter covers

- The types of data typically used by digital twins
- Sources of data and how they are integrated into a digital twin
- Data storage solutions
- Managing data governance and compliance

Data is the lifeblood of a digital twin. Sensors, enterprise systems, external APIs, and human inputs all generate data that must come together in a coherent way if the twin is to reflect reality and provide actionable insights. Without careful attention to how data is collected, cleaned, combined, and stored, a digital twin risks becoming fragmented, inconsistent, or untrustworthy. Data integration and management are therefore not just technical necessities but the foundation upon which a successful digital twin is built.

We've examined how to construct digital representations of physical systems and use sensors to gather data about system changes and feed it into your digital twin. Now we look at the ways in which we can integrate data into the twin from the myriad of places it may be produced, and how we can store and manage this data so that is available in an accurate and timely manner for the twin to consume. Together, these capabilities are what allow a digital twin to evolve from a collection of raw signals into a trusted representation that supports decision-making, prediction, and optimization.

We begin by looking at the different types of data that are important in a digital twin. You'll learn how each type plays a role in shaping the twin, what challenges they pose, and how they can be combined effectively. By the end of this section, you will be able to classify digital twin data sources, understand their unique characteristics, and recognize how they influence integration strategies. This foundation sets the stage for looking at how to store, transform, and govern this data so that your digital twin remains consistent, reliable, and useful over time.

4.1 Types of data

Digital twins must handle diverse data types—both structured and unstructured—that each bring their own characteristics, storage demands, and processing requirements. Understanding these different data formats is essential when designing a digital twin architecture capable of capturing, storing, and analyzing the complex information needed to create accurate representations of physical systems.